CLASS 10 · CBSE · NCERT MATHEMATICS · NCERT SOLUTIONS

Chapter 2: Polynomials

Step-by-step solutions to the textbook exercises: 8 questions

Questions are numbered as in the textbook. Try each one yourself before reading the solution.

1 Exercise 2.1

Q1 Six graphs of $y = p(x)$ are given in the textbook (Fig. 2.10). Find the number of zeroes of $p(x)$ in each case.

SOLUTION

The zeroes of $p(x)$ are the x-coordinates of the points where the graph of $y = p(x)$ meets the x-axis. So count the points where each graph cuts or touches the x-axis.

Graph	What the graph does	Number of zeroes
(i)	A straight line parallel to the x-axis, above it. It never meets the x-axis.	0
(ii)	A curve that stays below the x-axis on the left and crosses it once on the right.	1
(iii)	A curve that crosses the x-axis at three points.	3
(iv)	A U-shaped curve that crosses the x-axis at two points.	2
(v)	A wavy curve that crosses the x-axis at four points.	4
(vi)	A curve that crosses the x-axis once and touches it at two more points.	3

Answer: (i) 0 (ii) 1 (iii) 3 (iv) 2 (v) 4 (vi) 3

2 Exercise 2.2

Q1 (i) Find the zeroes of $x^2 - 2x - 8$ and verify the relationship between the zeroes and the coefficients.

SOLUTION

$x^2 - 2x - 8 = x^2 - 4x + 2x - 8 = x(x-4) + 2(x-4) = (x-4)(x+2)$. The zeroes are 4 and -2.

Sum of zeroes $= 4 + (-2) = 2 = \dfrac{-(-2)}{1} = -\dfrac{\text{coefficient of } x}{\text{coefficient of } x^2}$.

Product of zeroes $= 4 \times (-2) = -8 = \dfrac{-8}{1} = \dfrac{\text{constant term}}{\text{coefficient of } x^2}$.

Answer: zeroes 4 and -2. Verified.

Q1 (ii) Find the zeroes of $4s^2 - 4s + 1$ and verify the relationship between the zeroes and the coefficients.

SOLUTION

$4s^2 - 4s + 1 = (2s - 1)^2$. Both zeroes are $\frac{1}{2}$.

Sum of zeroes $= \frac{1}{2} + \frac{1}{2} = 1 = \frac{-(-4)}{4}$.

Product of zeroes $= \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} = \frac{\text{constant term}}{\text{coefficient of } s^2}$.

Answer: zeroes $\frac{1}{2}$ and $\frac{1}{2}$. Verified.

Q1 (iii) Find the zeroes of $6x^2 - 3 - 7x$ and verify the relationship between the zeroes and the coefficients.

SOLUTION

In standard form the polynomial is $6x^2 - 7x - 3 = 6x^2 - 9x + 2x - 3 = 3x(2x - 3) + 1(2x - 3) = (3x + 1)(2x - 3)$. The zeroes are $-\frac{1}{3}$ and $\frac{3}{2}$.

Sum of zeroes $= -\frac{1}{3} + \frac{3}{2} = \frac{7}{6} = \frac{-(-7)}{6}$.

Product of zeroes $= -\frac{1}{3} \times \frac{3}{2} = -\frac{1}{2} = \frac{-3}{6}$.

Answer: zeroes $-\frac{1}{3}$ and $\frac{3}{2}$. Verified.

Q1 (iv) Find the zeroes of $4u^2 + 8u$ and verify the relationship between the zeroes and the coefficients.

SOLUTION

$4u^2 + 8u = 4u(u + 2)$. The zeroes are 0 and -2.

Sum of zeroes $= 0 + (-2) = -2 = \frac{-8}{4}$.

Product of zeroes $= 0 \times (-2) = 0 = \frac{0}{4}$, since the constant term is 0.

Answer: zeroes 0 and -2. Verified.

Q1 (v) Find the zeroes of $t^2 - 15$ and verify the relationship between the zeroes and the coefficients.

SOLUTION

$t^2 - 15 = (t - \sqrt{15})(t + \sqrt{15})$. The zeroes are $\sqrt{15}$ and $-\sqrt{15}$.

Sum of zeroes $= \sqrt{15} - \sqrt{15} = 0 = \frac{-0}{1}$, since the coefficient of t is 0.

Product of zeroes $= \sqrt{15} \times (-\sqrt{15}) = -15 = \frac{-15}{1}$.

Answer: zeroes $\sqrt{15}$ and $-\sqrt{15}$. Verified.

Q1 (vi) Find the zeroes of $3x^2 - x - 4$ and verify the relationship between the zeroes and the coefficients.

SOLUTION

$3x^2 - x - 4 = 3x^2 + 3x - 4x - 4 = 3x(x+1) - 4(x+1) = (3x-4)(x+1)$. The zeroes are $\frac{4}{3}$ and -1.

Sum of zeroes $= \frac{4}{3} - 1 = \frac{1}{3} = \frac{-(-1)}{3}$.

Product of zeroes $= \frac{4}{3} \times (-1) = -\frac{4}{3} = \frac{-4}{3}$.

Answer: zeroes $\frac{4}{3}$ and -1. Verified.

Q2 Find a quadratic polynomial whose zeroes have the given sum and product: (i) $\frac{1}{4}, -1$ (ii) $\sqrt{2}, \frac{1}{3}$ (iii) $0, \sqrt{5}$ (iv) $1, 1$ (v) $-\frac{1}{4}, \frac{1}{4}$ (vi) $4, 1$.

SOLUTION

A quadratic polynomial with sum of zeroes S and product of zeroes P is $x^2 - Sx + P$. Any non-zero multiple of it has the same zeroes, so fractions can be cleared.

(i) $x^2 - \frac{1}{4}x - 1$. Multiplying by 4: $4x^2 - x - 4$.

(ii) $x^2 - \sqrt{2}\,x + \frac{1}{3}$. Multiplying by 3: $3x^2 - 3\sqrt{2}\,x + 1$.

(iii) $x^2 - 0x + \sqrt{5} = x^2 + \sqrt{5}$.

(iv) $x^2 - x + 1$.

(v) $x^2 + \frac{1}{4}x + \frac{1}{4}$. Multiplying by 4: $4x^2 + x + 1$.

(vi) $x^2 - 4x + 1$.

Answer: (i) $4x^2 - x - 4$ (ii) $3x^2 - 3\sqrt{2}\,x + 1$ (iii) $x^2 + \sqrt{5}$ (iv) $x^2 - x + 1$ (v) $4x^2 + x + 1$ (vi) $x^2 - 4x + 1$

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